First week of development : Ovulation to implantation.

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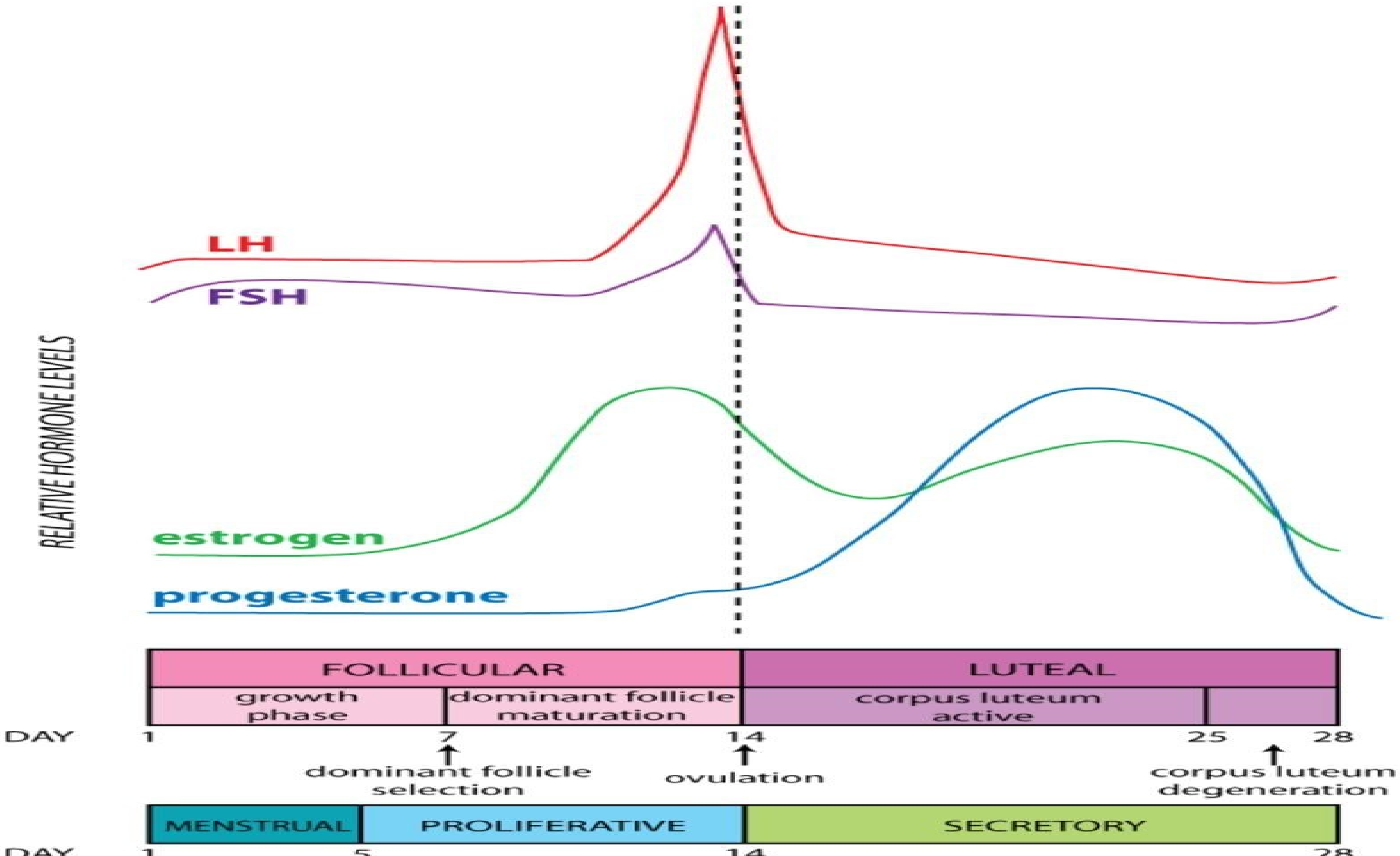
Ovarian Cycle:

1) Control axis (who commands who)

- Hypothalamus releases GnRH
- GnRH stimulates anterior pituitary → releases FSH and LH
- FSH/LH drive cyclic ovarian changes

Ovarian Cycle (cont.):

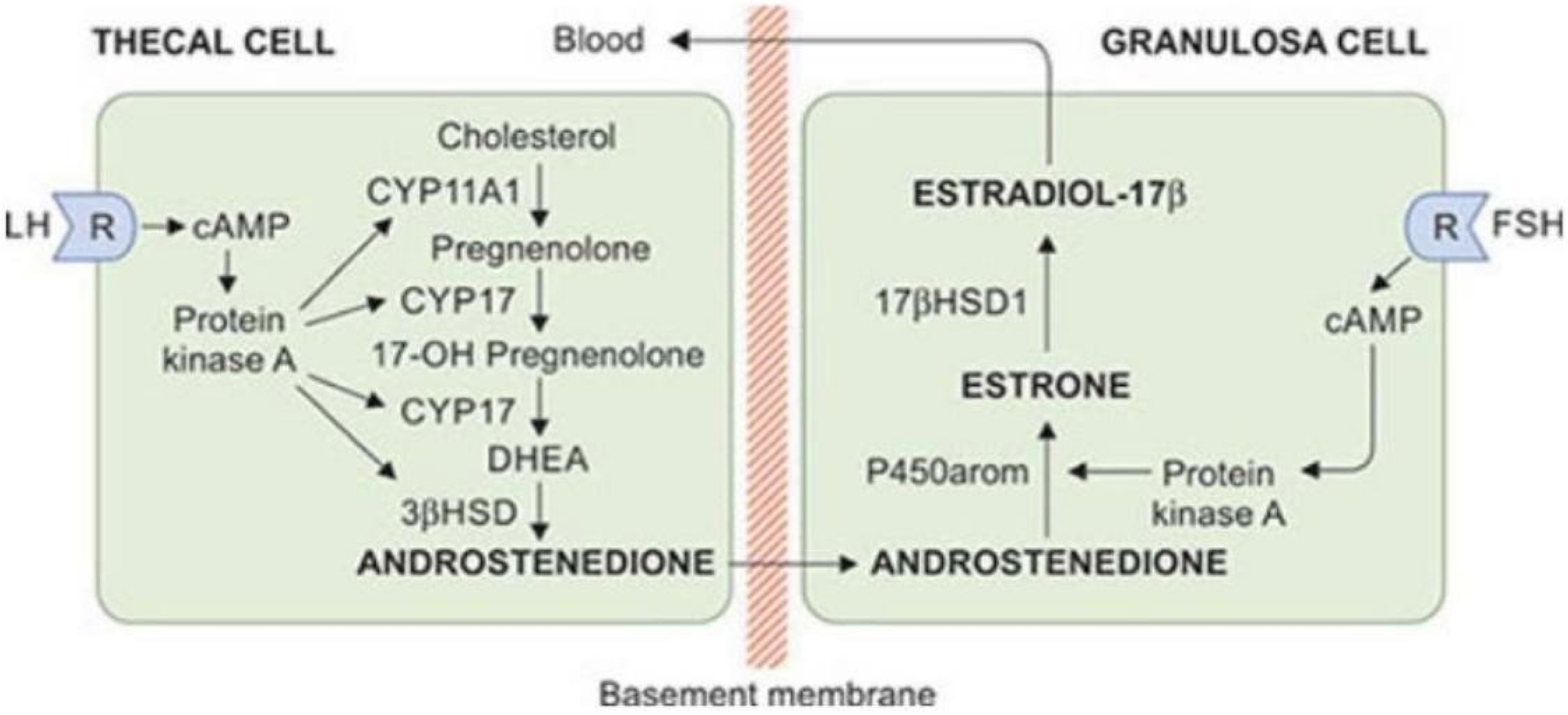
- **2) Early follicular phase: recruitment + selection(days 1-14)**
- At the start of each cycle, FSH “rescues” ~15–20 primary (preantral) follicles to grow.
- Without enough FSH support, follicles undergo atresia.
- Typically only 1 follicle becomes dominant and reaches full maturity → 1 oocyte is ovulated.
- The rest degenerate; after atresia, tissue is replaced by connective tissue → corpus atreticum.


LH
FSH
RELATIVE HORMONE LEVELS
estrogen
progesterone
FOLLICULAR
LUTEAL
growth phase
dominant follicle maturation
corpus luteum active
DAY
1
7
14
25
28
dominant follicle selection
ovulation
corpus luteum degeneration
MENSTRUAL
PROLIFERATIVE
SECRETORY
DAY
1
5
14
28

Ovarian Cycle (cont.):

- **3) Estrogen production: “two-cell, two-gonadotropin” idea**
- LH acts on theca interna → makes androgens (androstenedione, testosterone).
- FSH acts on granulosa cells → converts androgens → estrogens (estrone, 17β-estradiol).

Armstrong-Dorrington “two-cell” model



Ovarian Cycle (cont.):

- **4) Main effects of rising estrogen (follicular/proliferative phase)**
- Endometrium enters proliferative (follicular) phase
- Cervical mucus thins → easier sperm passage
- High estrogen eventually promotes LH secretion → sets up the surge

Ovarian Cycle (cont.):

5) Midcycle: LH surge (the trigger)LH surge causes:

- Oocyte completes meiosis I and starts meiosis II
- Luteinization of follicular cells → ↑ progesterone production
- Follicular rupture → ovulation

6) Ovulation

- Just before ovulation, under FSH + LH, the follicle rapidly enlarges to a mature (Graafian) follicle (~25 mm diameter).

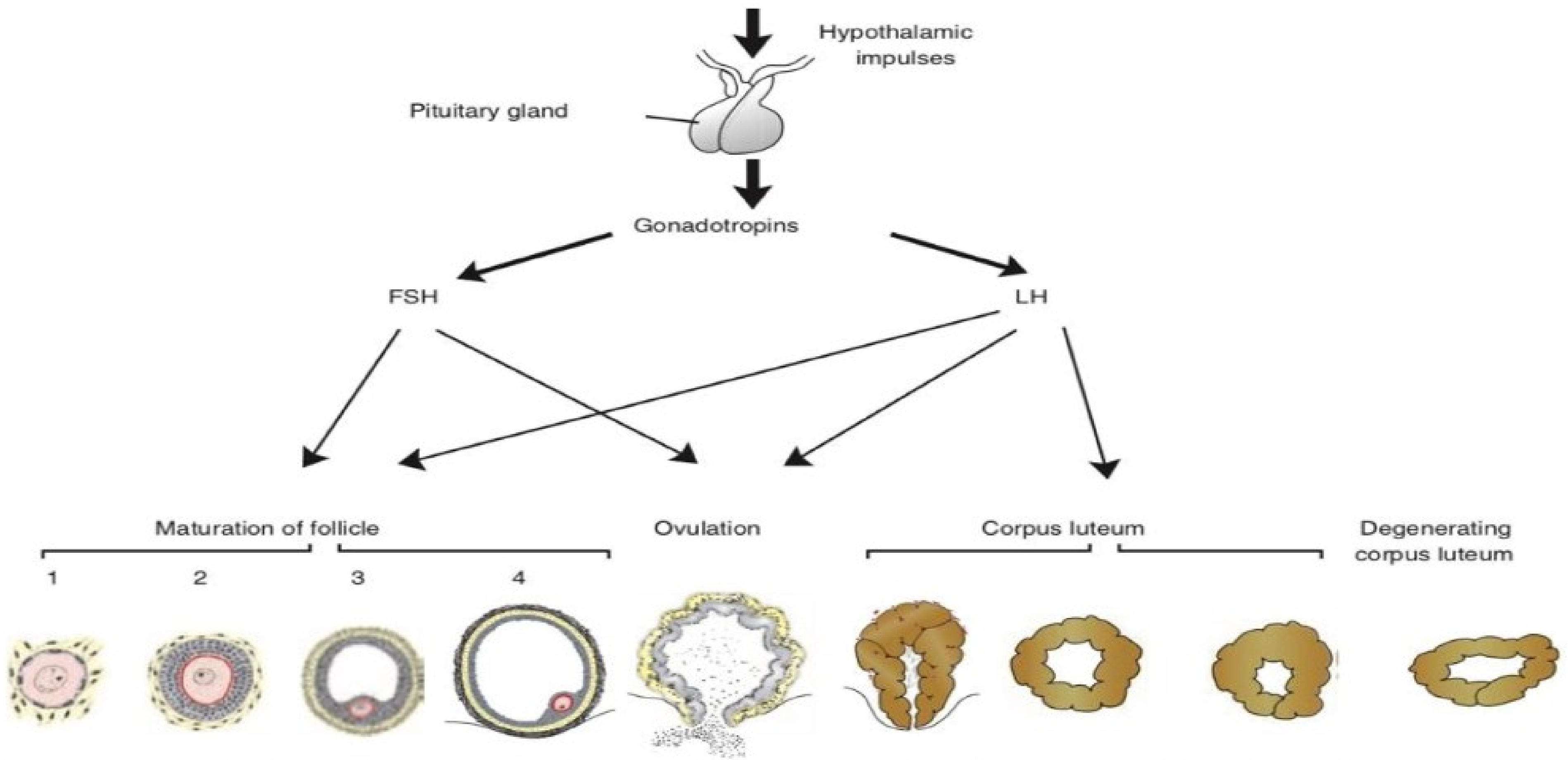


Figure 3.1 Drawing showing the role of the hypothalamus and pituitary gland in regulating the ovarian cycle. Under the influence of GnRH from the hypothalamus, the pituitary releases the gonadotropins, FSH, and LH. Follicles are stimulated to grow by FSH and to mature by FSH and LH. Ovulation occurs when concentrations of LH surge to high levels. LH also promotes development of the corpus luteum. *1*, primordial follicle; *2*, growing follicle; *3*, vesicular follicle; *4*, mature vesicular (graafian) follicle.

Clinical correlates:

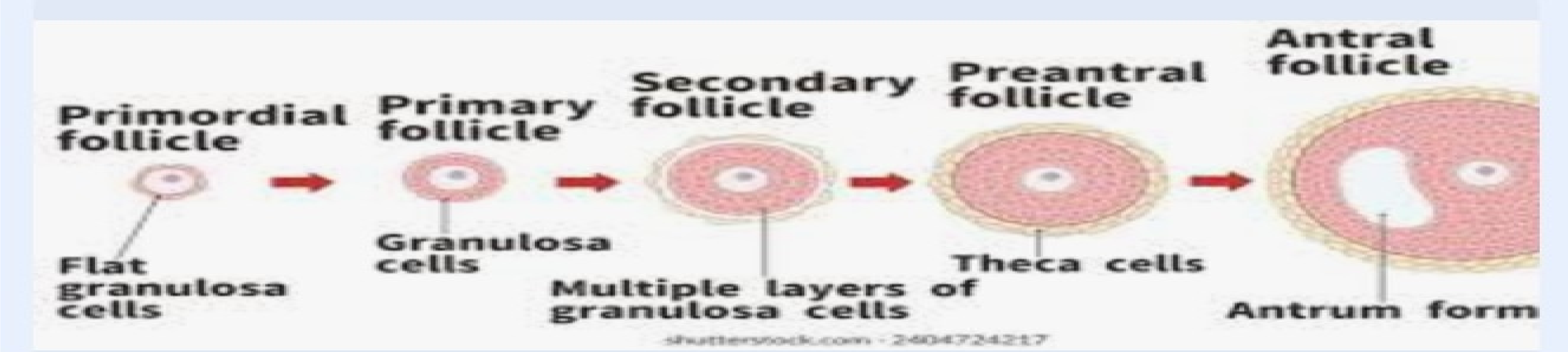
- Mittelschmerz: midcycle unilateral pelvic pain
- Basal body temperature rises after ovulation (progesterone effect)
- Low gonadotropins → anovulation; ovulation induction meds can cause multiple ovulations → higher multiple pregnancy rate.

Ovulation:

- Follicle → Ovulation → Corpus luteum
- 1) Follicle growth (Primordial → Growing → Vesicular/Graafian)
- Primordial follicle: primary oocyte + single layer of flattened follicular cells.
- Growing follicle: granulosa cells proliferate; zona pellucida forms; theca interna/externa develop.
- Vesicular (antral/Graafian) follicle: fluid-filled antrum appears; follicle enlarges and becomes hormone-active.

Ovulation (cont.):

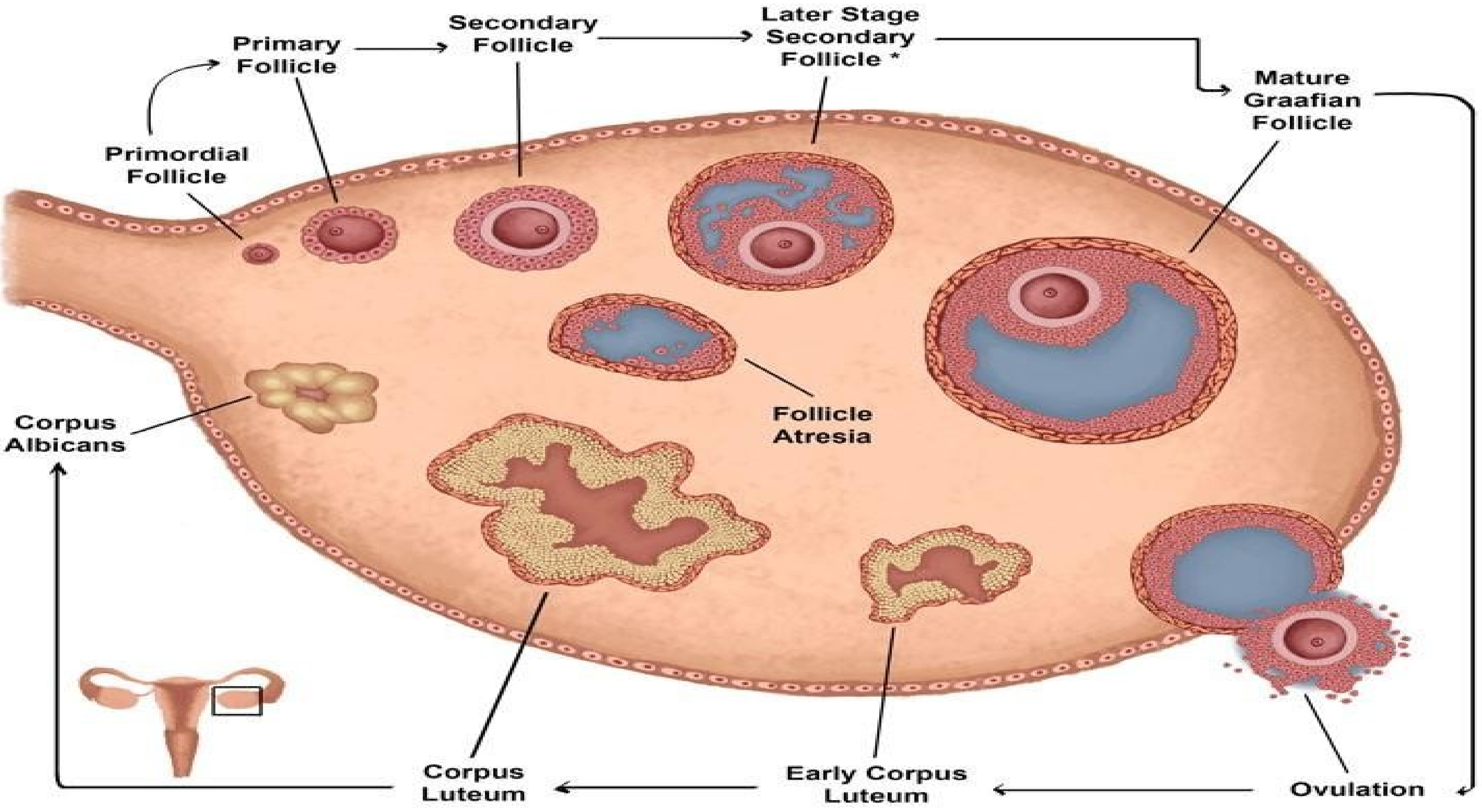
- 2) Hormonal control
- FSH recruits and supports follicle growth (especially granulosa).
- Rising estrogen (from granulosa + theca cooperation) triggers LH surge.
- LH surge → final follicle maturationn + ovulation.



Ovulation (cont.):

- 3) What is released at ovulation
- The follicle ruptures and releases the secondary oocyte (arrested in metaphase II) surrounded by corona radiata/cumulus cells.
- 4) Corpus luteum
- After rupture, remaining granulosa + theca interna cells luteinize → corpus luteum.
- Secretes mainly progesterone (and some estrogen) → converts endometrium to secretory (implantation-ready) phase.

*Starting at the later stage for secondary follicles, an antrum begins to develop. Follicles from this point on can be referred to as "antral follicles"



Ovulation (cont.):

- A mature follicle ruptures → releases a secondary oocyte surrounded by a cloud of cells (cumulus oophorus).
- The fimbriae (finger-like ends of the uterine tube) sweep over the ovary and catch the oocyte.
- How the oocyte moves inside the uterine tube (oocyte transport)Once inside the tube, movement is driven mainly by:
- 1. Cilia (tiny hair-like “fans” on the tubal lining)
- 2. Peristalsis (smooth muscle “waves” squeezing the tube)

Where fertilization usually happens?

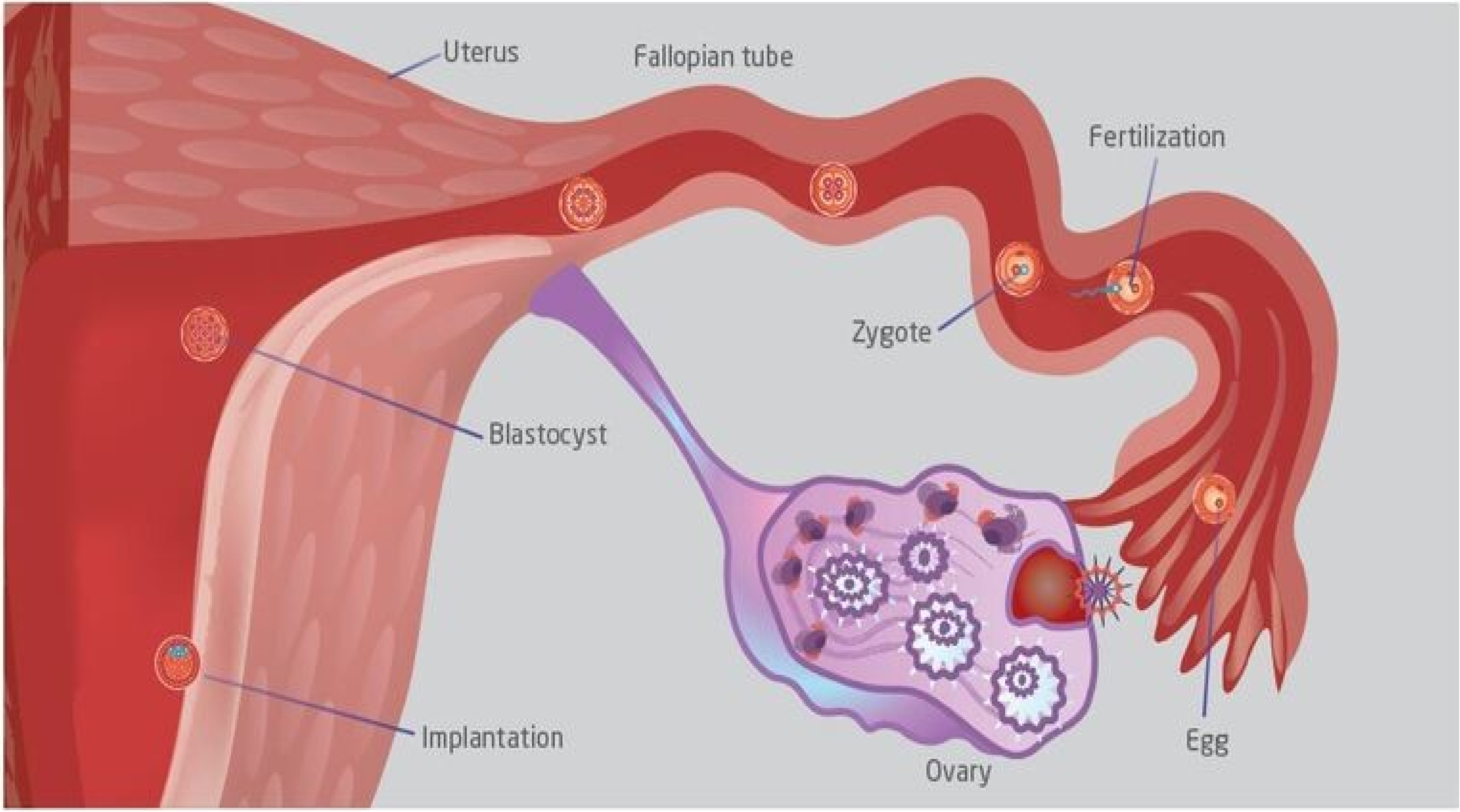
- Fertilization most commonly occurs in the ampulla of the uterine tube (the wider, outer part near the ovary).Why ampulla? It’s the widest part and is positioned where the oocyte arrives soon after ovulation.
- What happens to the follicle after ovulation ?
- After the follicle ruptures, the remaining follicle becomes the corpus luteum (“yellow body”).
- Main job: secrete progesterone (and some estrogen) to prepare and maintain the endometrium.

If fertilization does not occur?

- Corpus luteum peaks ~9 days after ovulation, then regresses → forms a fibrous scar called the corpus albicans.
- Progesterone falls → menstrual bleeding occurs

If fertilization does occur?

- The early embryo produces hCG (human chorionic gonadotropin), which rescues the corpus luteum so it keeps making progesterone.
- This becomes the corpus luteum of pregnancy, supporting progesterone production until the placenta can take over (classically around the end of the first trimester


Uterus
Fallopian tube
Fertilization
Zygote
Blastocyst
Implantation
Ovary
Egg

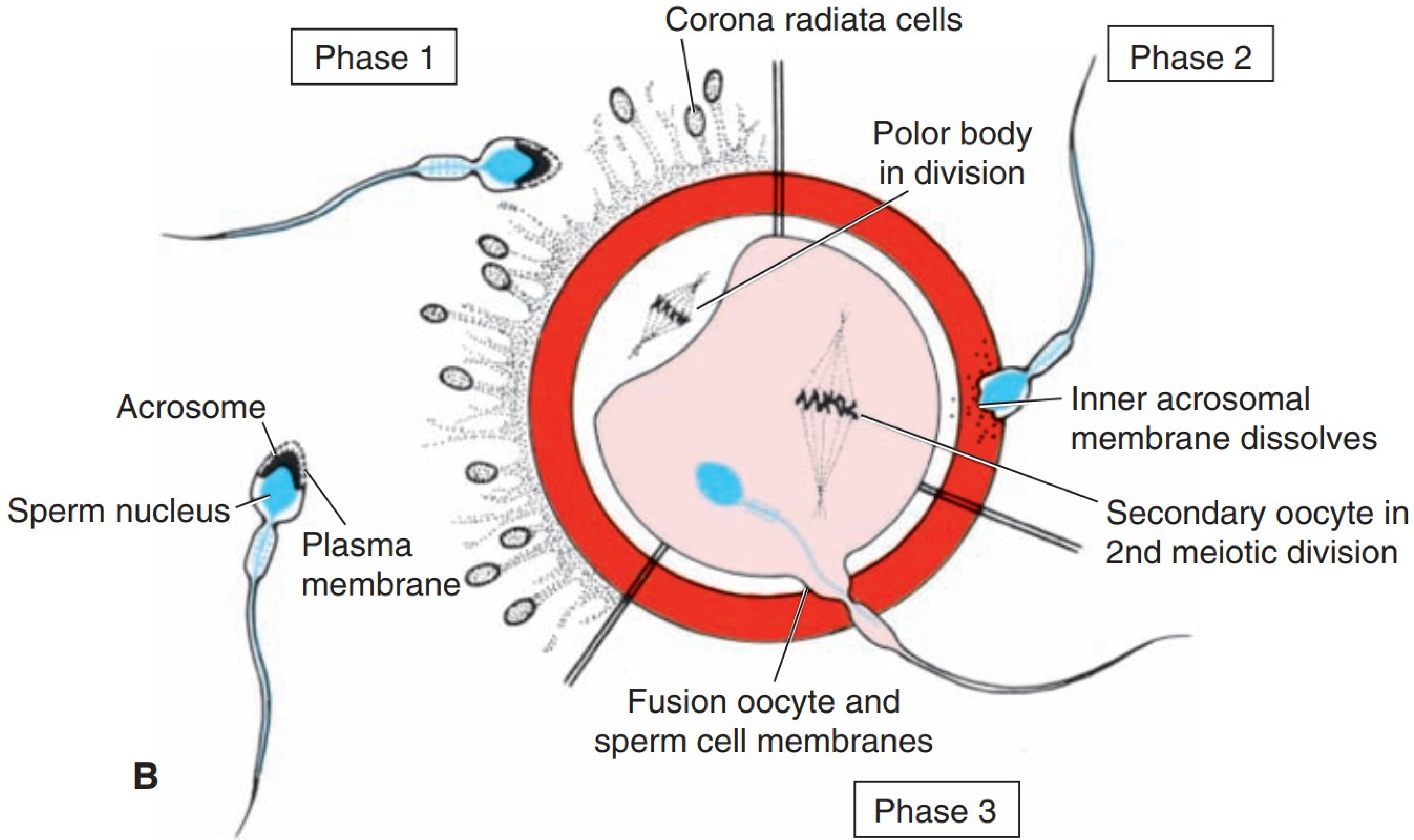
Fertilization

The process by which male and female gametes fuse, occurs in the ampullary region of the uterine tube, this is the widest part of the tube and is close to the ovary.

Spermatozoa may remain viable in the female reproductive tract for several days. Only 1% of sperm deposited in the vagina enter the cervix, where they may survive for many hours. Movement of sperm from the cervix to the uterine tube occurs by muscular contractions of the uterus and uterine tube and very little by their own propulsion.

Fertilization (cont.):

- Spermatozoa are not able to fertilize the oocyte immediately upon arrival in the female genital tract but must undergo:
- (1) capacitation and (2) the acrosome reaction to acquire this capability.
- **Capacitation** is a period of conditioning in the female reproductive tract that in the human lasts approximately 7 hours.
- Removes surface “coating” (glycoproteins/seminal plasma proteins) → sperm membrane becomes more responsive.
- Result: sperm becomes able to:
- 1. bind properly to the egg coverings
- 2. do the acrosome reaction


Phase 1
Corona radiata cells
Phase 2
Polor body
in division
Acrosome
Sperm nucleus
Plasma
membrane
Inner acrosomal
membrane dissolves
Secondary oocyte in
2nd meiotic division
Fusion oocyte and
sperm cell membranes
B
Phase 3

Fertilization (cont.):

- **The acrosome reaction**, which occurs after binding to the zona pellucida, is induced by zona proteins. This reaction culminates in the release of enzymes needed to penetrate the zona pellucida, including acrosin- and trypsin-like substances.
- **The phases of fertilization include**
- ● Phase 1, penetration of the corona radiata
- ● Phase 2, penetration of the zona pellucida
- ● Phase 3, fusion of the oocyte and sperm cell membranes.

What happens in the oocyte immediately after sperm enters (3 responses):

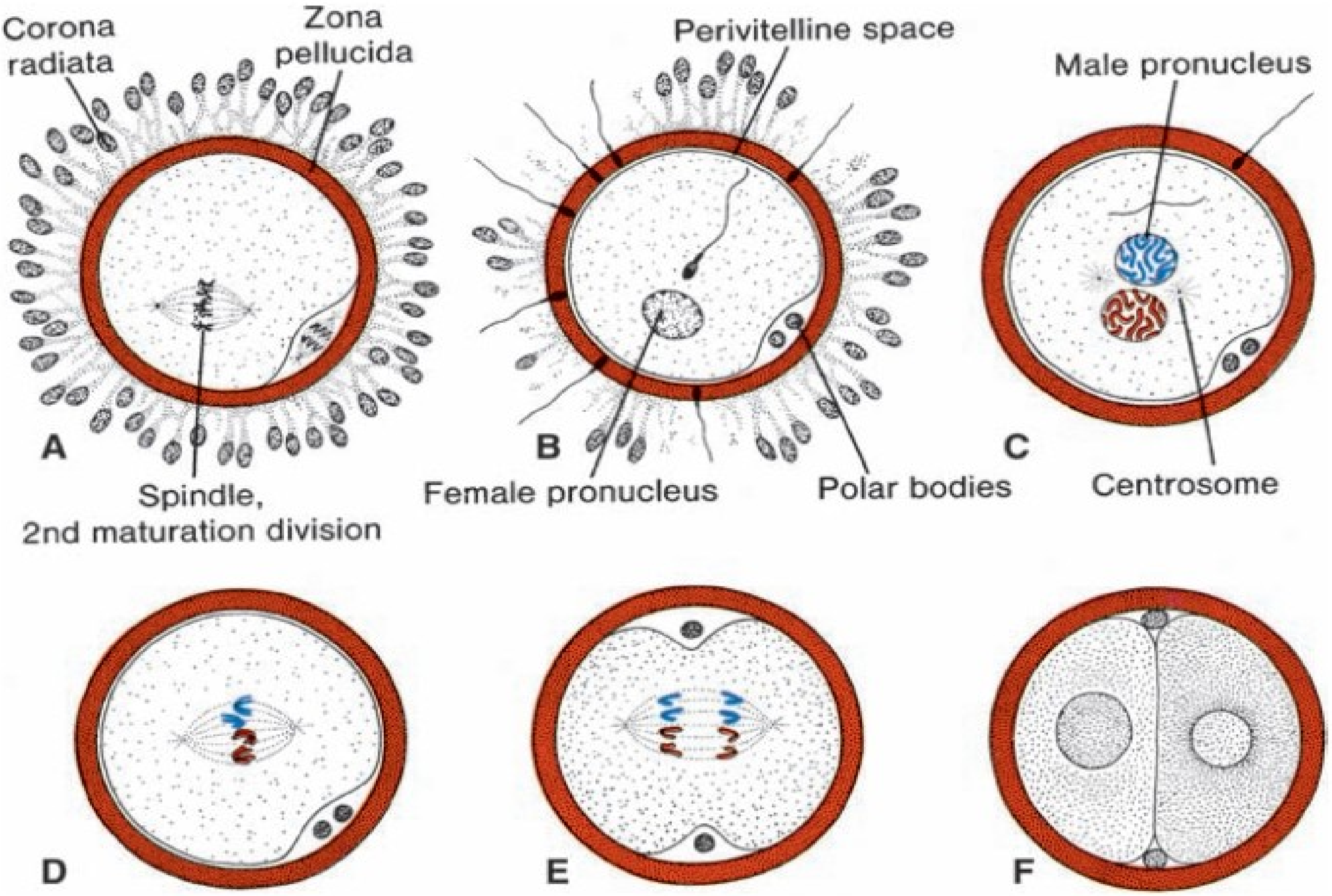
- **1) Cortical reaction + zona reaction (block polyspermy)**
- Cortical granules release enzymes → oocyte membrane becomes resistant to more sperm.
- Zona pellucida changes structure → sperm can't bind/penetrate again.
- Purpose: prevents polyspermy (more than one sperm fertilizing the egg).

What happens in the oocyte immediately after sperm enters (3 responses) cont.:

- **2) Completion of meiosis II**
- The secondary oocyte (stuck in metaphase II) completes meiosis II → forms:
- definitive oocyte (ovum)
- second polar body
- Female chromosomes form the female pronucleus (haploid).
- **3) Metabolic activation of the egg**
- Egg ramps up metabolism and prepares for the first mitotic divisions.

Pronuclei and formation of the zygote

- **Male pronucleus**
- Sperm nucleus swells → becomes male pronucleus.
- Sperm tail degenerates.
- Sperm also contributes the centriole/centrosome (important for first mitotic spindle).
- **Female pronucleus** From the completed meiosis II products.


Corona radiata
Zona pellucida
Perivitelline space
Male pronucleus
A
B
C
Spindle, 2nd maturation division
Female pronucleus
Polar bodies
Centrosome
D
E
F

Pronuclei and formation of the zygote (cont.)

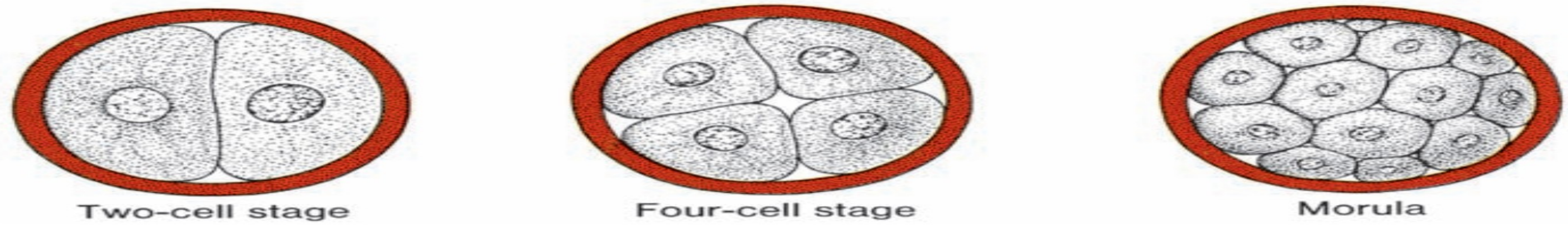
- Both pronuclei replicate DNA (still haploid).
- They come close, membranes break down, chromosomes align → first mitosis begins.
- Result = zygote with diploid (46) chromosomes.

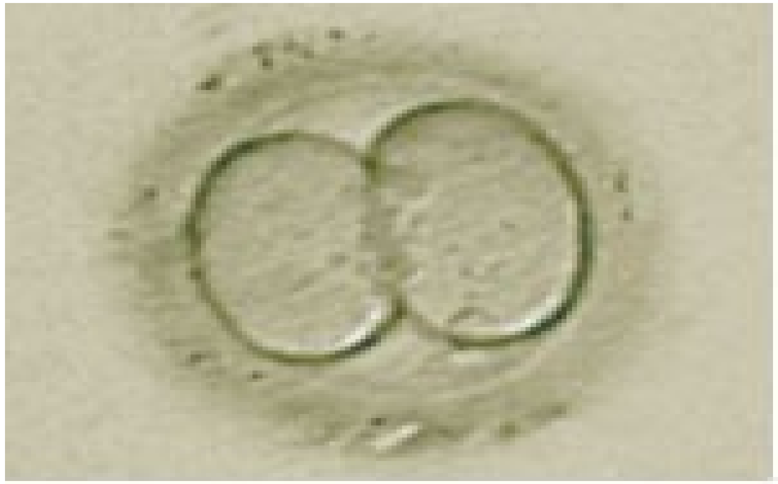
Cleavage:

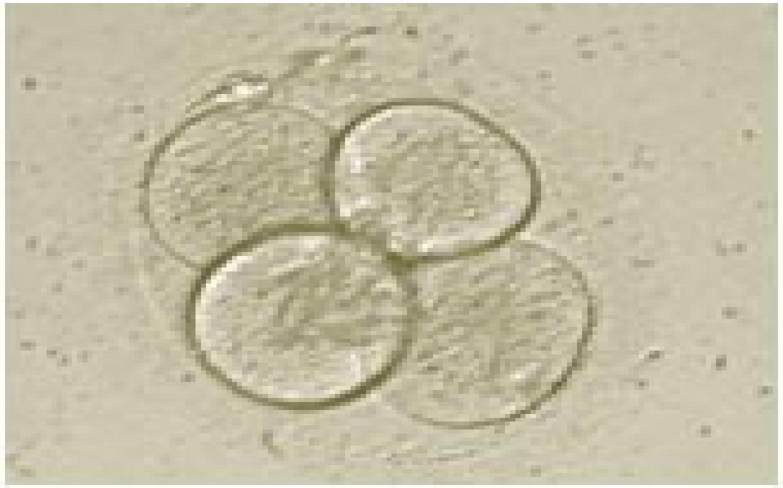
- Without fertilization, the oocyte degenerates soon after ovulation.
- Cleavage = rapid mitotic divisions that increase cell number without increasing total size (still inside the same “shell”).
- Blastomeres = the cells formed during cleavage.
- Zona pellucida stays around the early embryo initially (prevents premature attachment in the tube)?
- Cleavage (why cells increase but embryo doesn’t get bigger)?

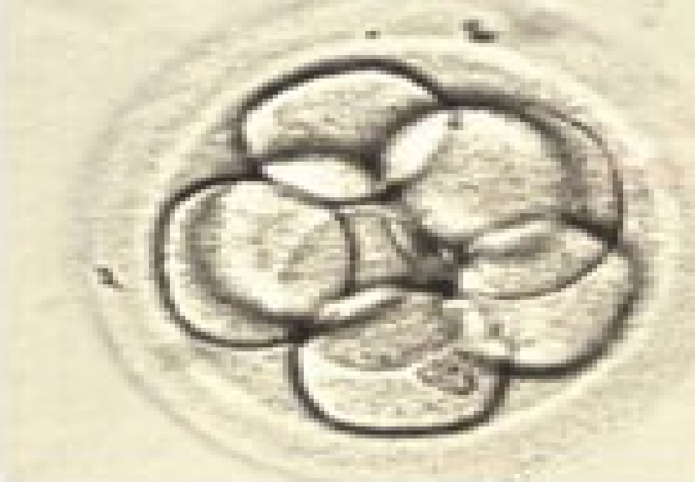
Cleavage (cont.):

- First cleavage \
- After fertilization, the zygote starts dividing:
- Two-cell stage is an early cleavage milestone.
- These early divisions happen while traveling down the tube toward the uterus.



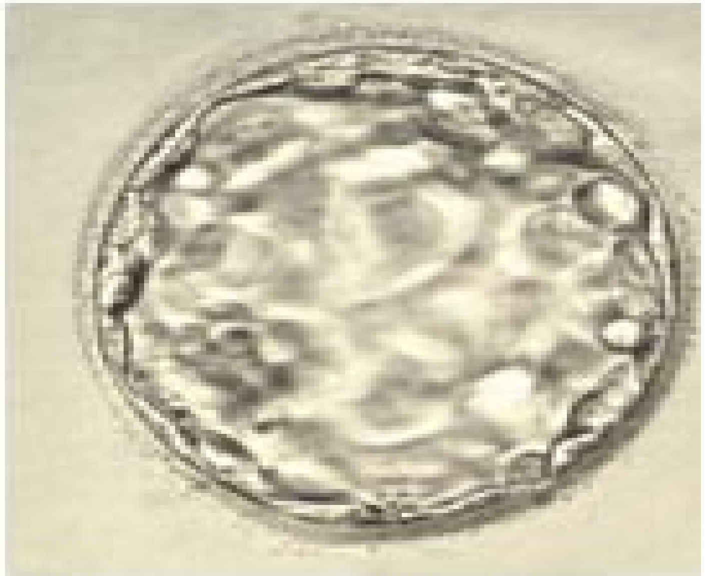

2-cell embryo


4-cell embryo


8-cell embryo


Morula


Early blastocyst

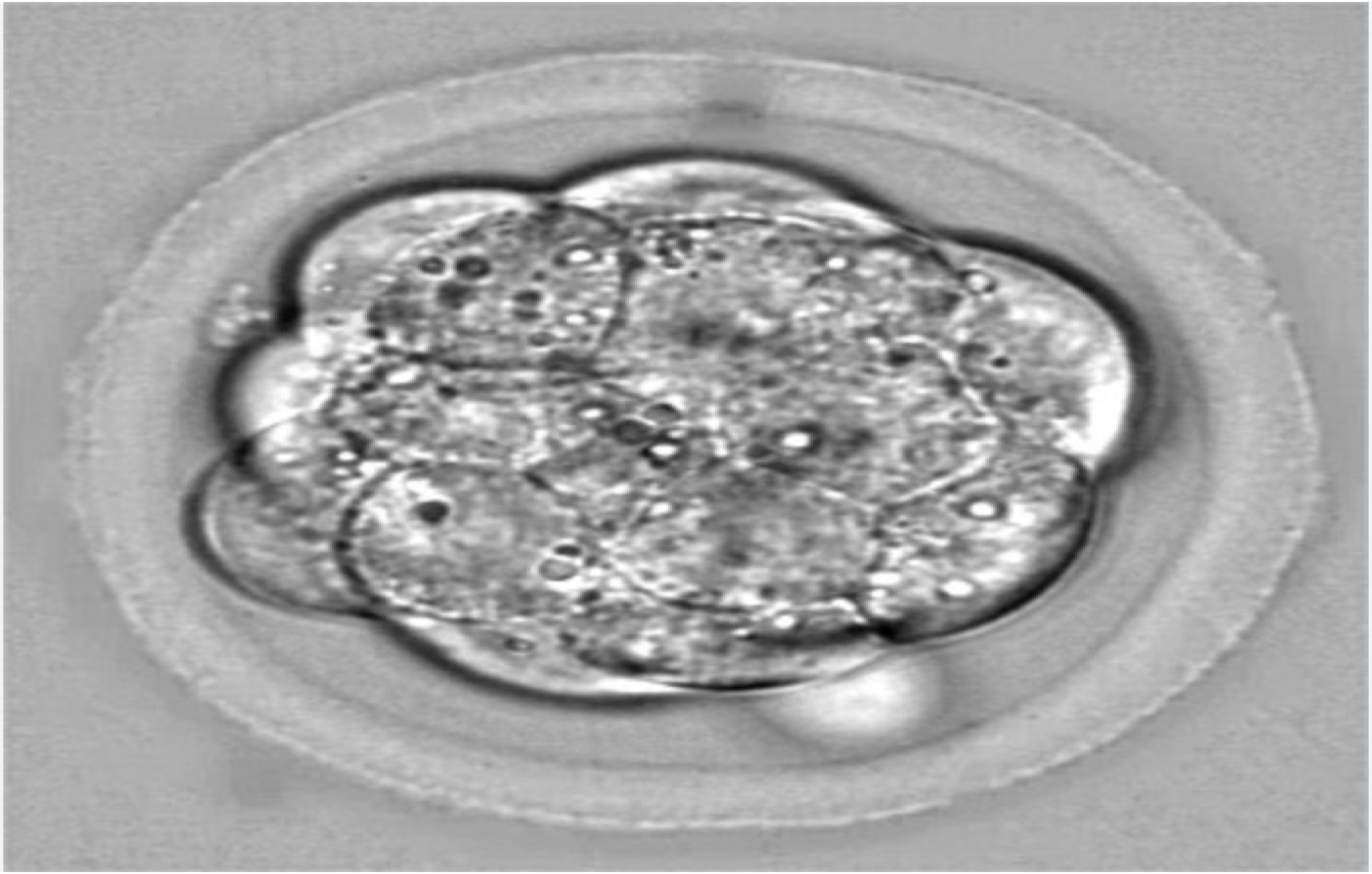

Expanded blastocyst

Compaction:

- After the third cleavage, however, blastomeres maximize their contact with each other, forming a compact ball of cells held together by tight junctions .
- At the 8-16 cells stage, blastomeres maximize contact and form tight junctions → embryo becomes a more solid, smooth ball.
- **Why it matters?**
- compaction helps separate cells into two populations:
- Outer cells → become trophoblast
- Inner cells → become inner cell mass (embryoblast)

Morula:

- Morula is the solid ball of cells formed from the early embryo after several cleavage divisions of the zygote.
- Stage: typically 12–16 cells (sometimes described up to ~32).
- Timing: around day 3 after fertilization.
- Location: usually still in the uterine tube and approaching/entering the uterus.
- Key feature: it’s solid (no cavity yet).

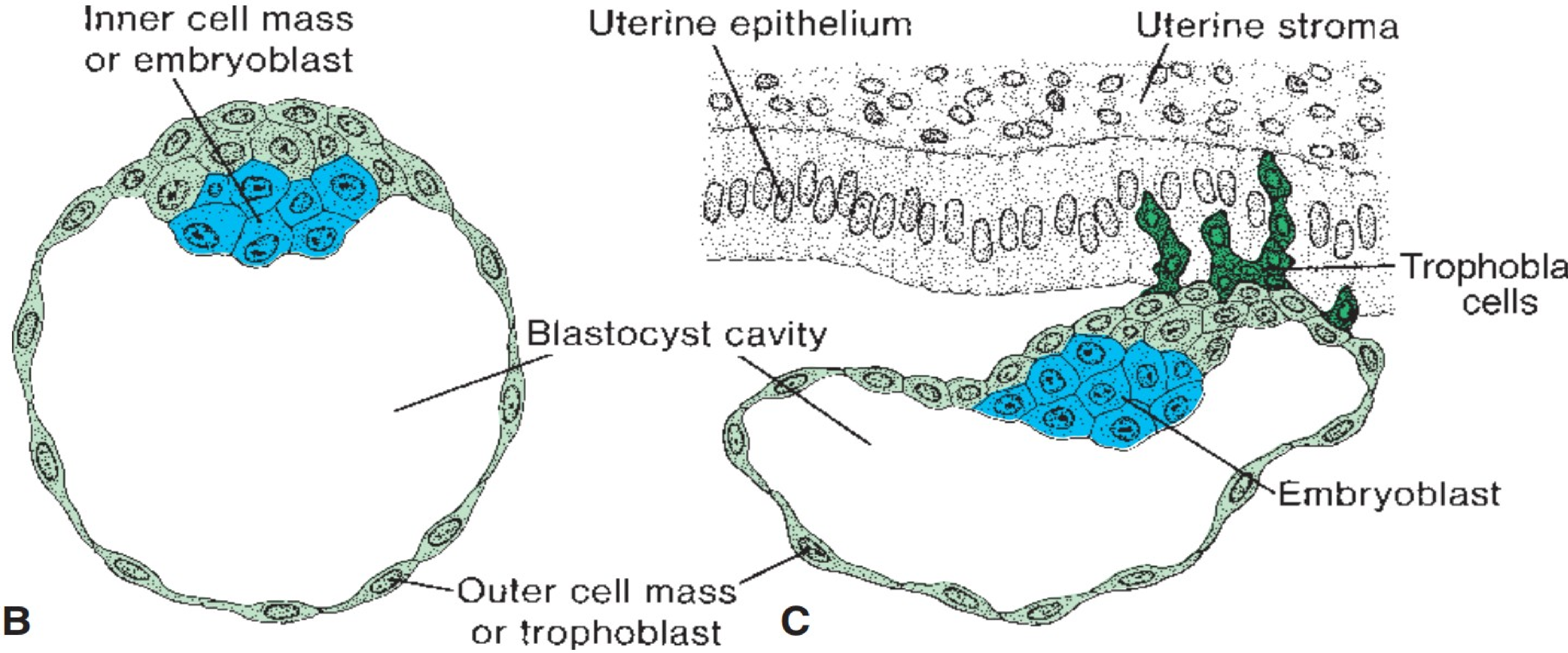


Blastocyst:

- As the morula enters the uterine cavity, fluid moves in, creating a cavity:
- Blastocyst cavity (blastocoel)Two main parts of a blastocyst (must-know)

1. Embryoblast (inner cell mass) → forms the embryo proper
2. Trophoblast (outer cell mass) → forms much of the placenta and is the first tissue to interact with endometrium.

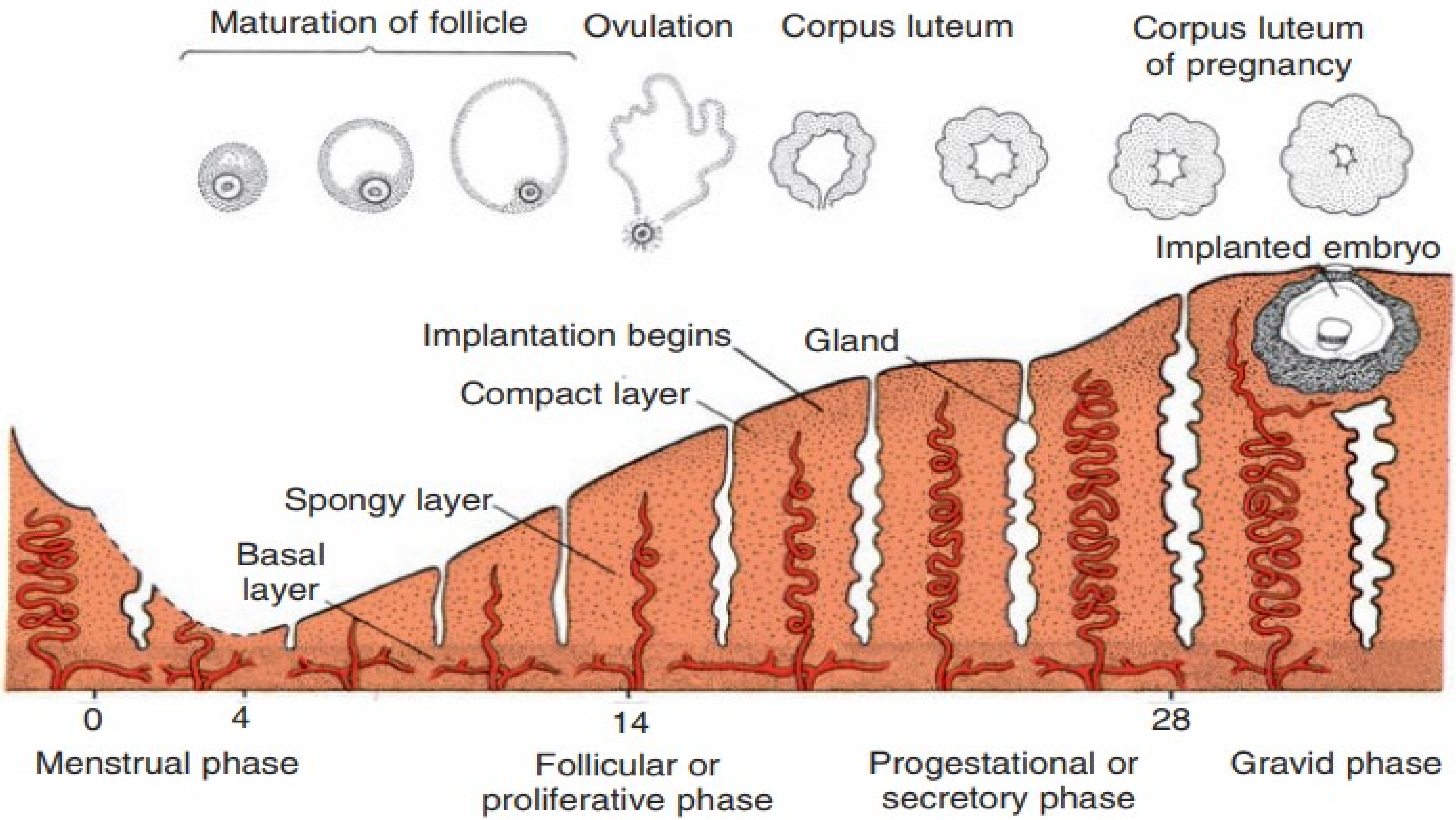
"Hatching" from the zona pellucida (needed for implantation)By about day 4–5, the zona pellucida disappears → blastocyst can expand and attach.



Inner cell mass
or embryoblast
Blastocyst cavity
Outer cell mass
or trophoblast
B
Uterine epithelium
Uterine stroma
Trophobla
cells
Embryoblast
C

Implantation:

- Implantation basics (start around day ~6)
- Implantation typically begins ~day 6 when trophoblast cells at the embryoblast pole contact uterine epithelium.
- The uterus is most receptive in the secretory (progesterone) phase.
- Uterus at implantation + endometrial layers
- Uterine wall layers
- 1.Endometrium (mucosa lining)
- 2.Myometrium (smooth muscle)
- 3.Perimetrium (outer covering)
- Endometrium divided into:
- Functional layer = builds up and sheds (menstruation)
- Basal layer = stays → regenerates endometrium after menses



Maturation of follicle
Ovulation
Corpus luteum
Corpus luteum of pregnancy
Implanted embryo
Implantation begins
Gland
Compact layer
Spongy layer
Basal layer
0
4
14
28
Menstrual phase
Follicular or proliferative phase
Progestational or secretory phase
Gravid phase

Endometrial phases (match to hormones):

- 1. Menstrual phase (shedding)
- 2. Proliferative/Follicular phase (estrogen → rebuilds endometrium)
- 3. Secretory/Luteal phase (progesterone → glands become active; best for implantation)